

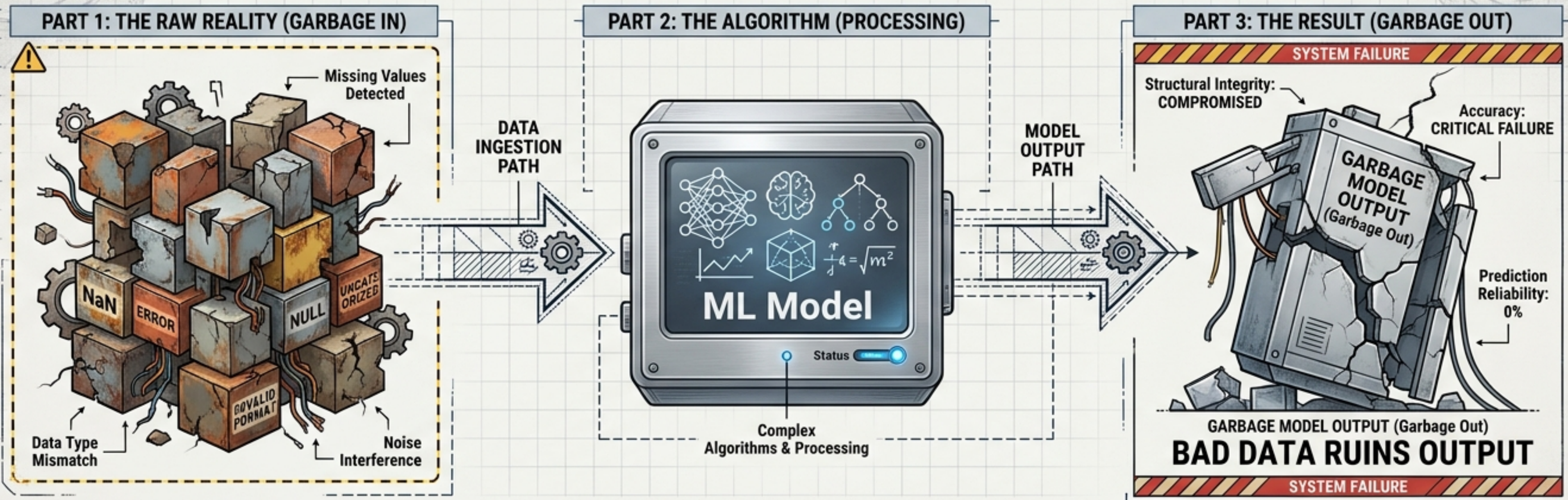
Pandas: Working With Real-World Data in Python

“The model isn’t bad. The dataset is just fighting for its life.” 🦴

AI Meme University — 15

THE DATA QUALITY LAW

AI DOES NOT START WITH A MODEL; IT STARTS WITH DATA.



THE REALITY

- Real datasets contain missing values.
- Wrong data formats and types.
- Irrelevant or noisy columns.



THE LAW

Garbage data → Garbage model.
A sophisticated algorithm cannot magically fix fundamentally bad data.

NO MAGIC FIX

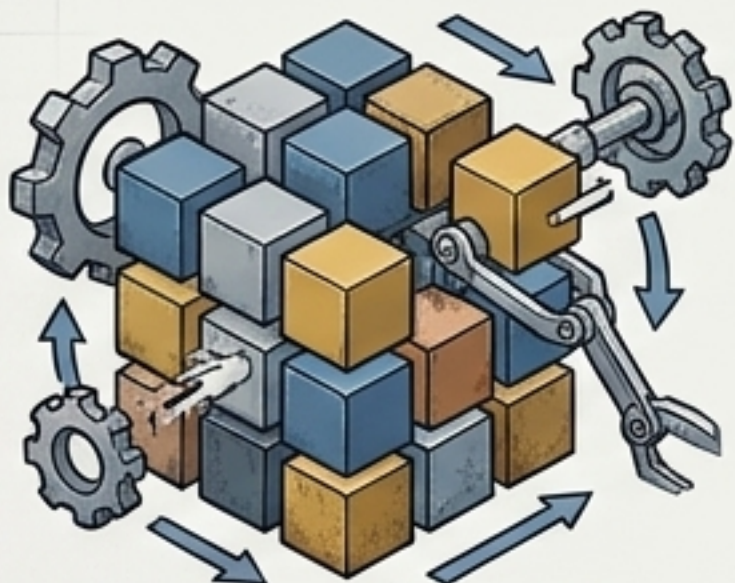
"Why is my model accuracy only 51%?"
→ Opens dataset
→ "Ah." 🦴

ENGINEER'S REALIZATION



Pandas is the command center for data manipulation.

Data Manipulation



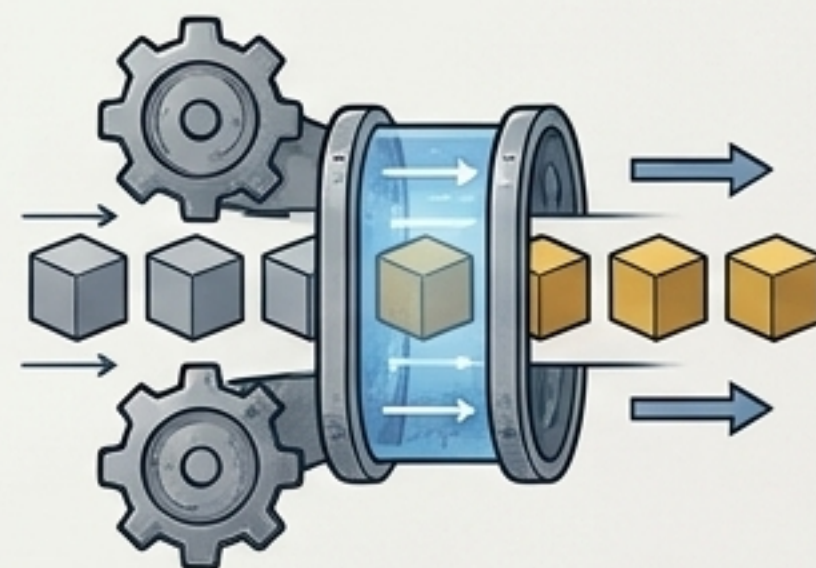
Data Analysis



Data Cleaning



Data Transformation

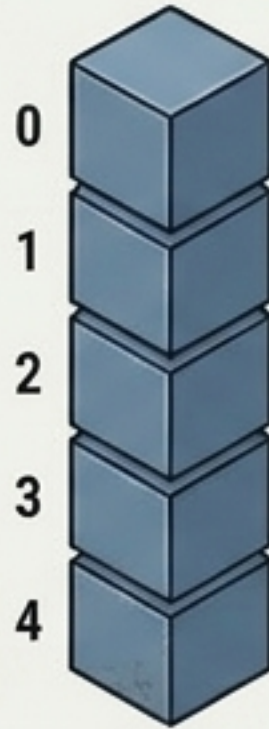


Pandas
Core

```
# The universal convention for importing Pandas  
import pandas as pd
```


THE TWO CORE STRUCTURES: SERIES VS. DATAFRAME

SERIES



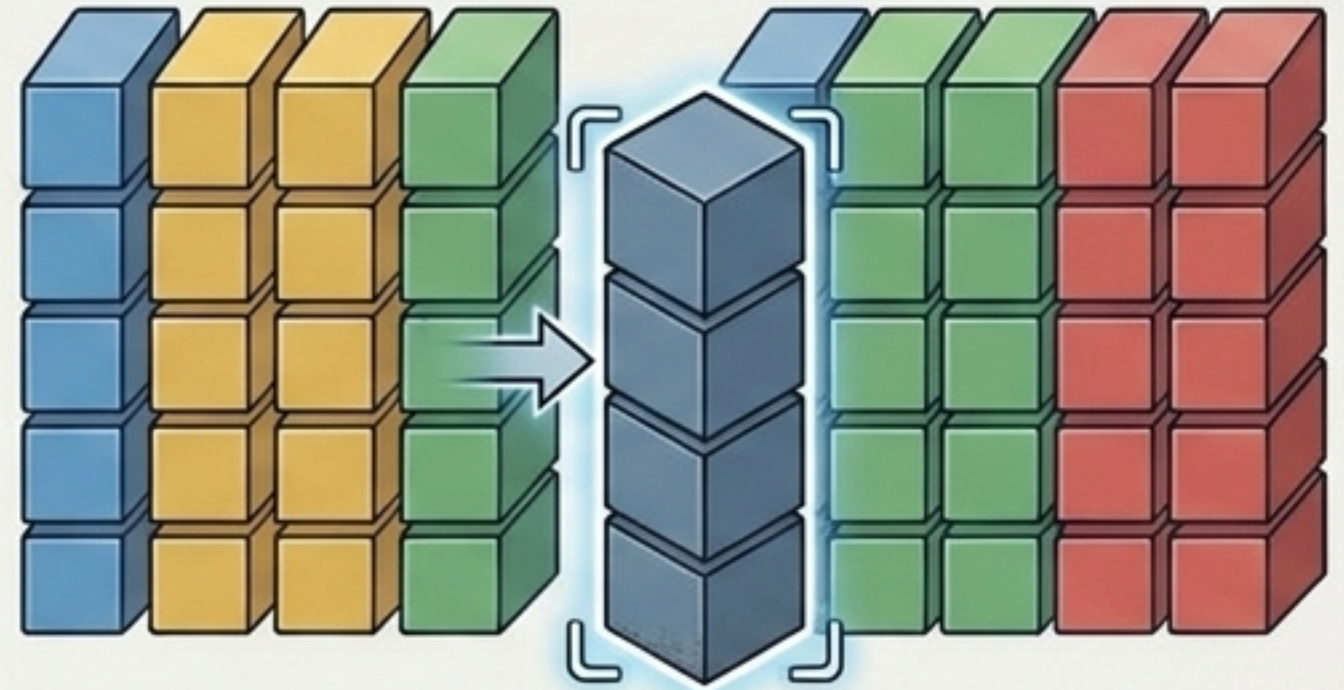
Concept: One-dimensional labeled collection.
Conceptually: One single column of data.

```
pd.Series([10, 20, 30])
```

```
0    10  
1    20  
2    30  
dtype: int64
```

COMPARISON MATRIX

DATAFRAME

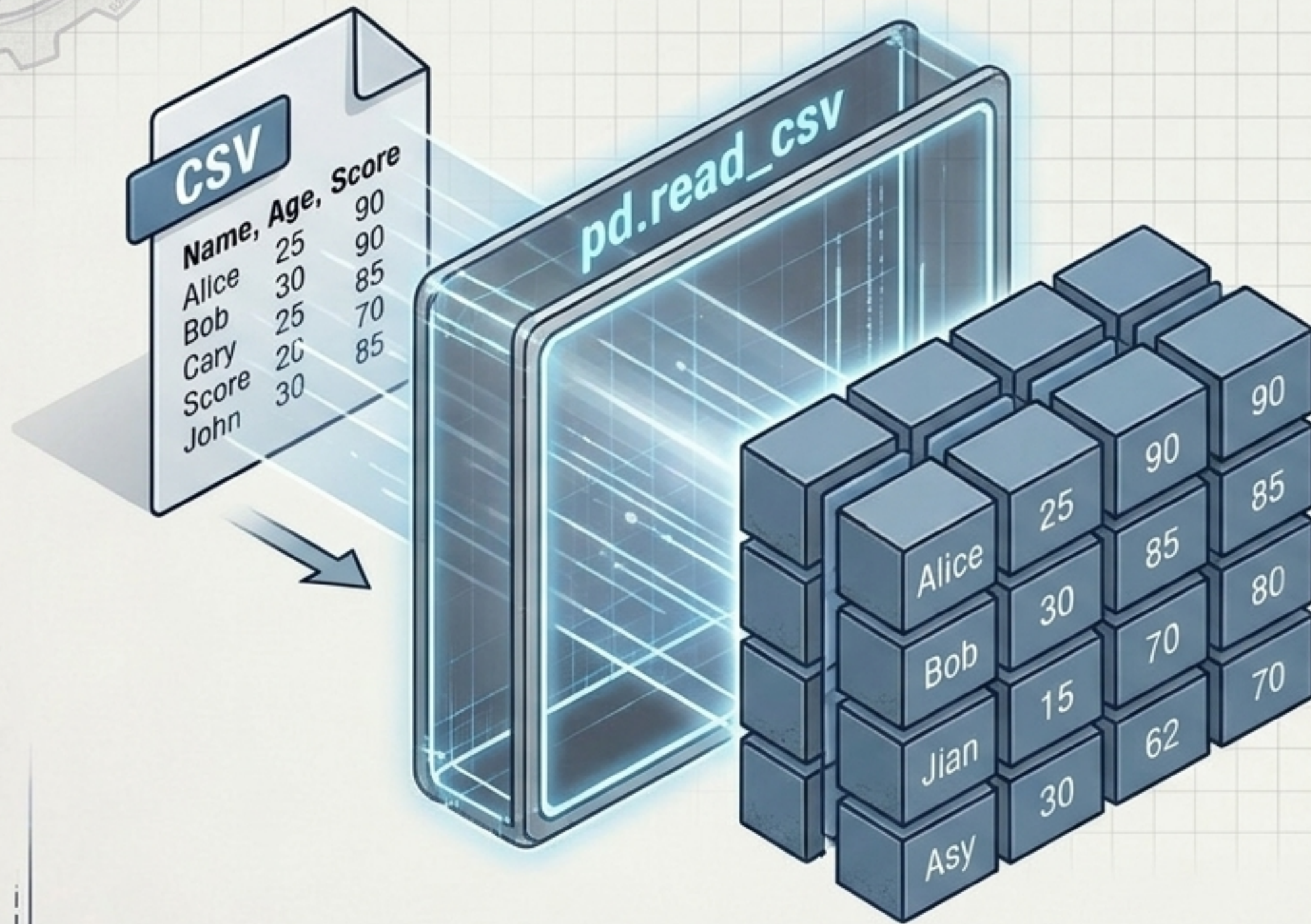


Concept: Two-dimensional tabular data.
Conceptually: A spreadsheet inside Python.

```
pd.DataFrame({'Name': ['Alice', 'Bob'],  
              'Age': [25, 30]})
```

	Name	Age
0	Alice	25
1	Bob	30

Ingestion: Reading structured datasets into Python



Input:

```
data.csv (Name, Age, Score)
```

Command:

```
df = pd.read_csv('data.csv')
```

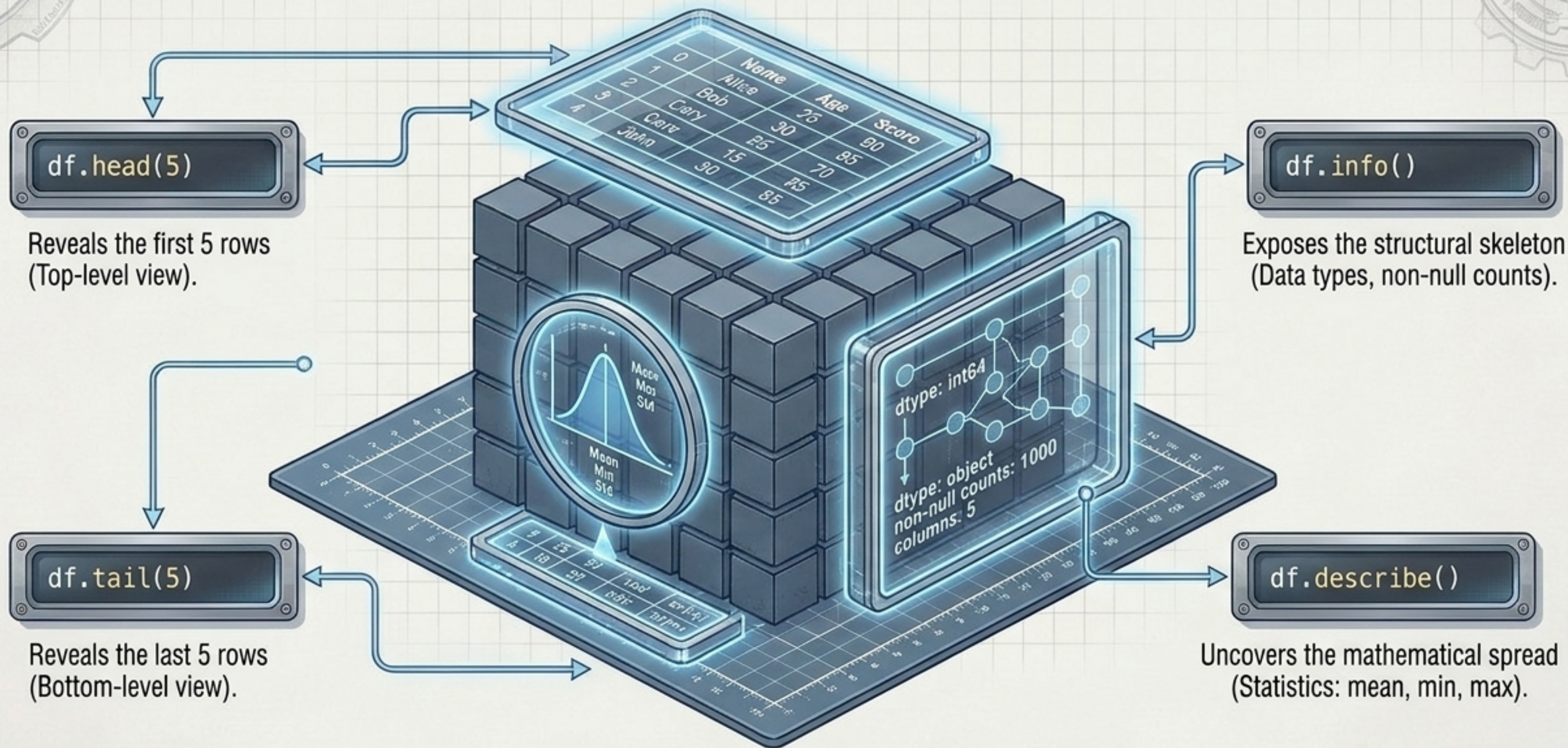
Result:

Pandas converts raw text into a fully manipulatable DataFrame object.

One of the most common dataset formats in real-world ML is the CSV (Comma Separated Values).

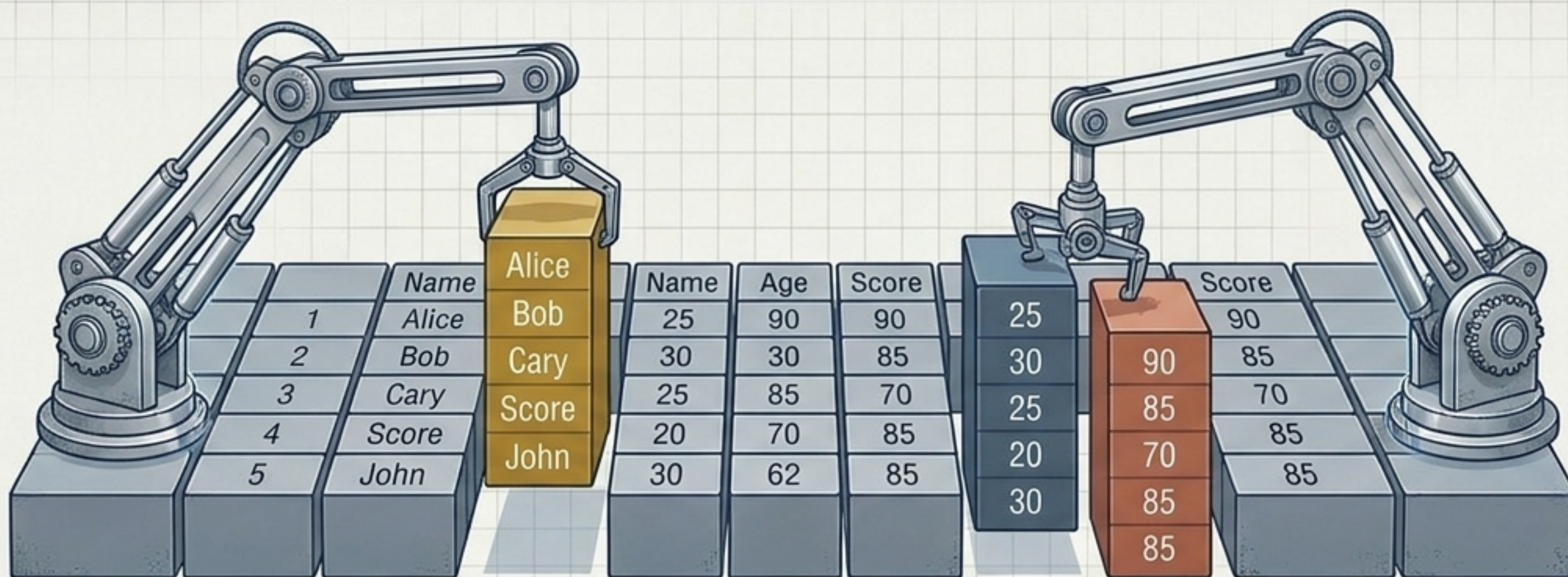


INSPECTION: Inspect the dataset before attempting any manipulation.



Inspecting your data structure, types, and statistical distribution is crucial before applying any transformations or models.

Selecting specific features isolates the data you need.



Select One Column (Returns a Series)

```
df['Name']
```

Isolating a single feature vector.

Select Multiple Columns (Returns a DataFrame)

```
df[['Name', 'Age']]
```

Isolating a subset of the dataset. Note the double brackets [[]].

Selecting the right features is a critical first step in Machine Learning to remove noise.

FILTERING: Filter datasets by applying logical row conditions.



```
# Returns only students whose score is greater than 80  
high_scorers = df[df['Score'] > 80]
```

Real-World Filtering Targets:



Age



Price



Category



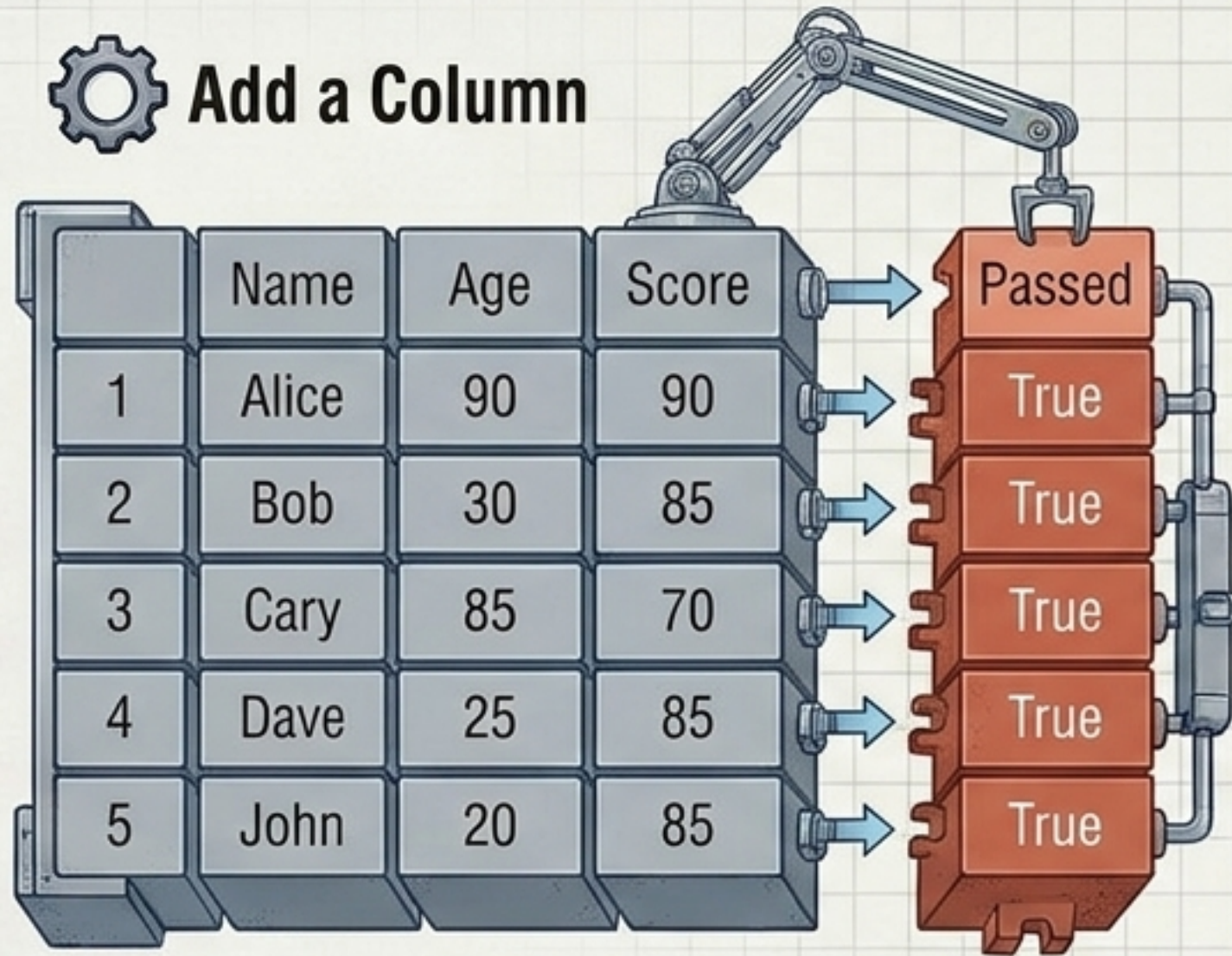
Location



Labels

Feature Engineering: Add and modify data columns.

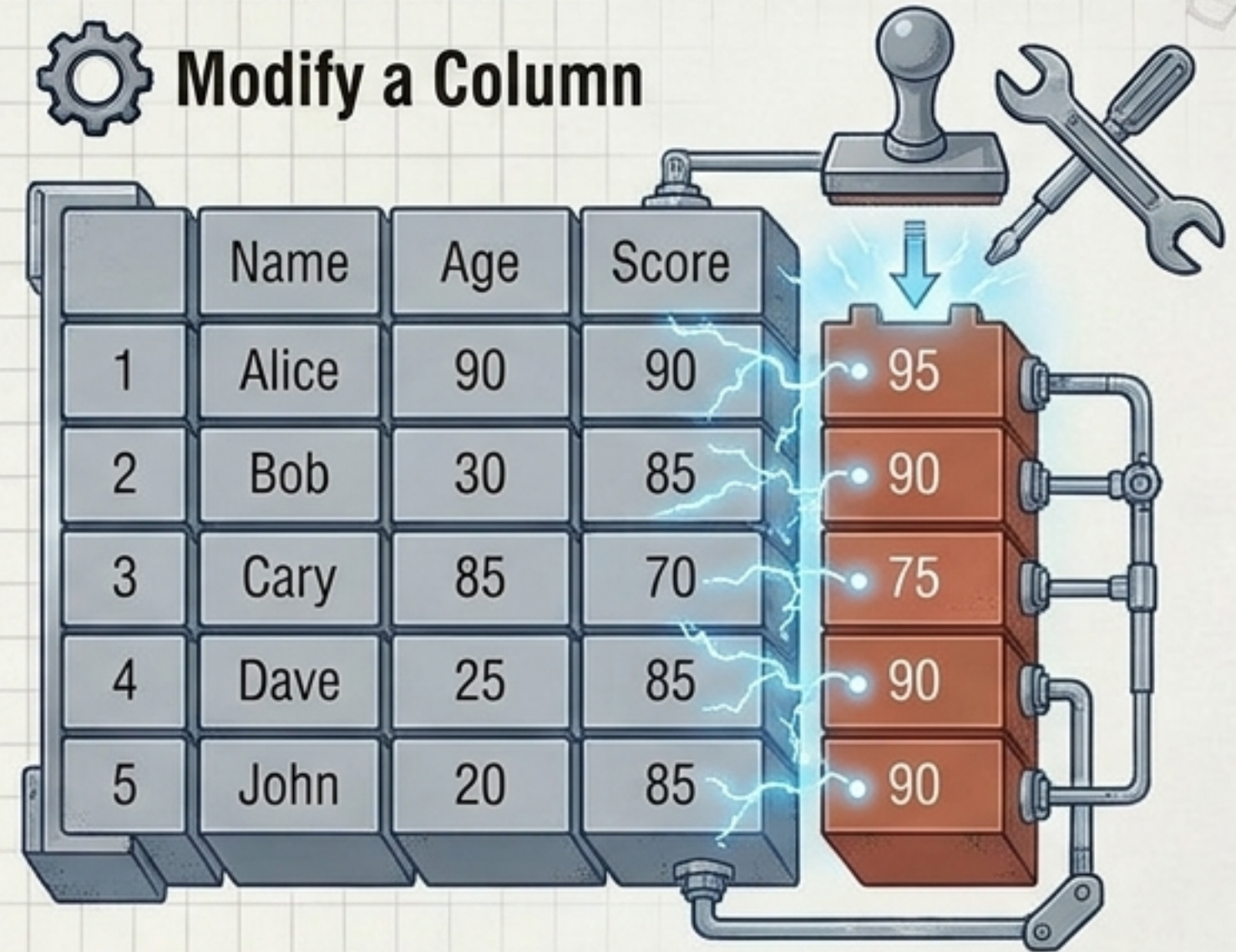
⚙️ Add a Column



```
df['Passed'] = True
```

Instantly expands the DataFrame's dimensions with a new feature.

⚙️ Modify a Column

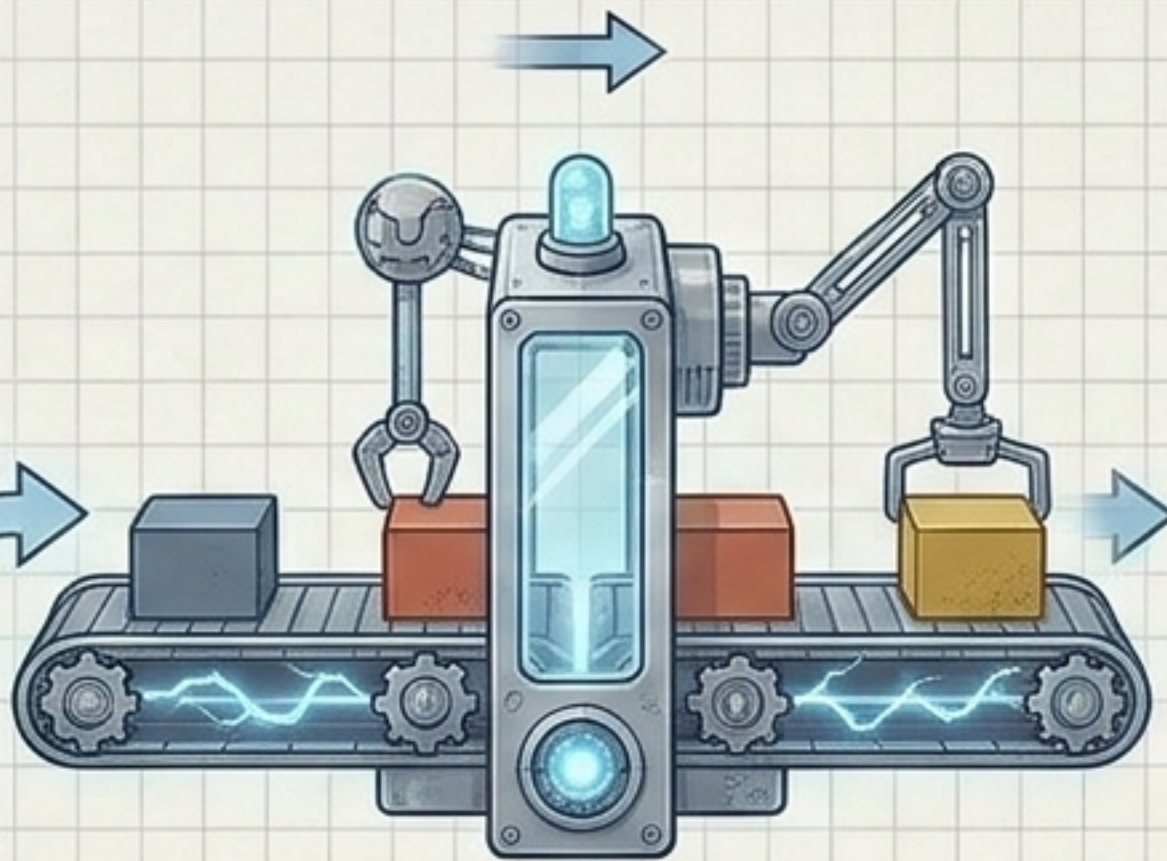
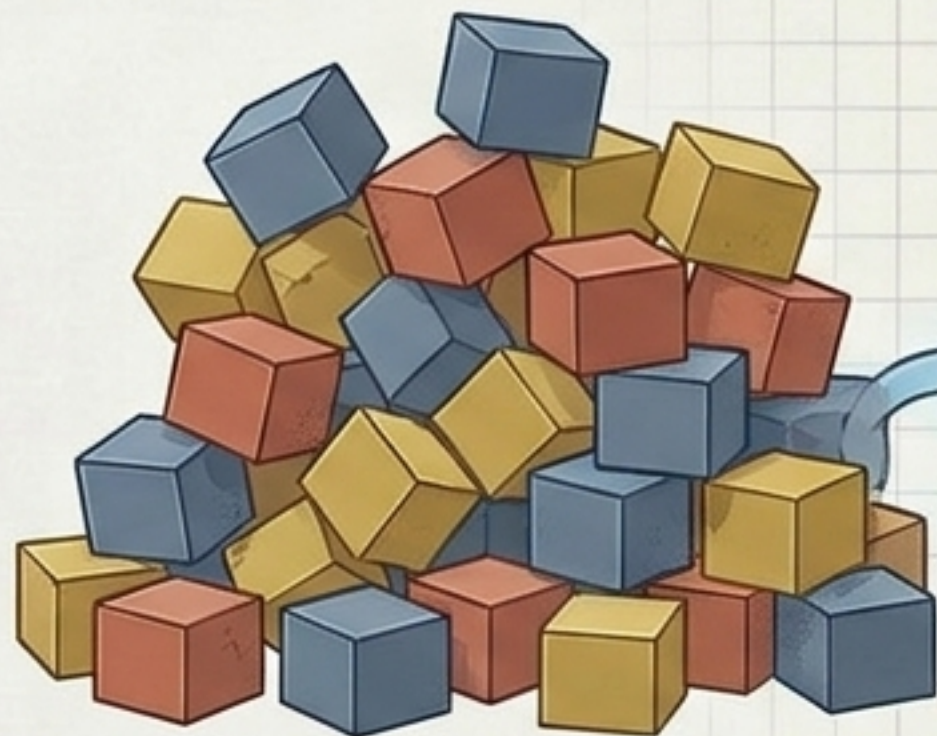


```
df['Score'] = df['Score'] + 5
```

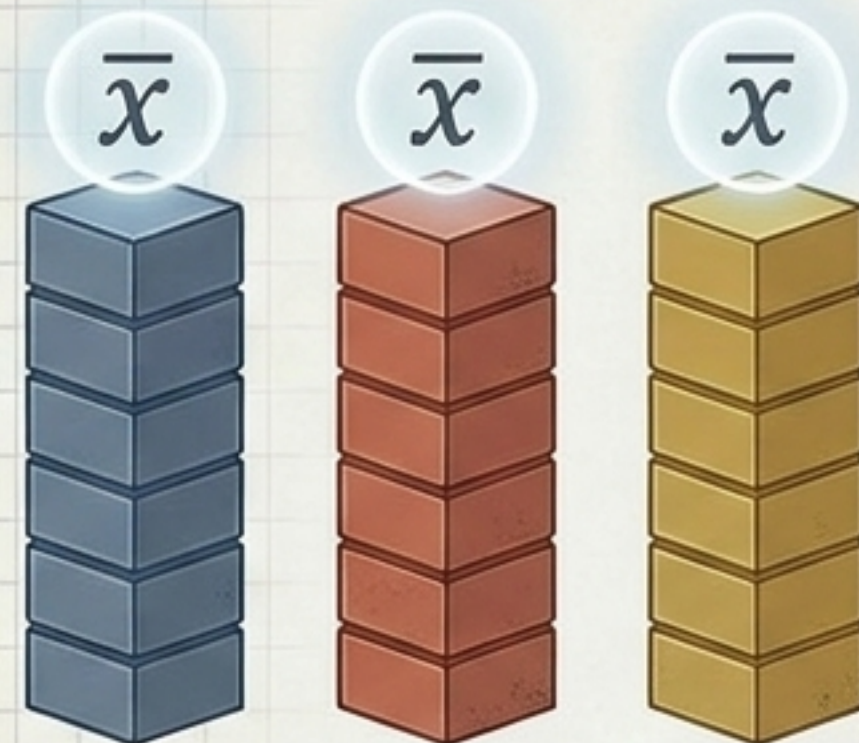
Applies mathematical transformations to every row in the column simultaneously.

Grouping and sorting reveals hidden patterns in the data.

Before



After



Sort (Order the data)

```
df.sort_values('Score', ascending=False)
```

Organizes rows in descending order.

Group (Aggregate the data)

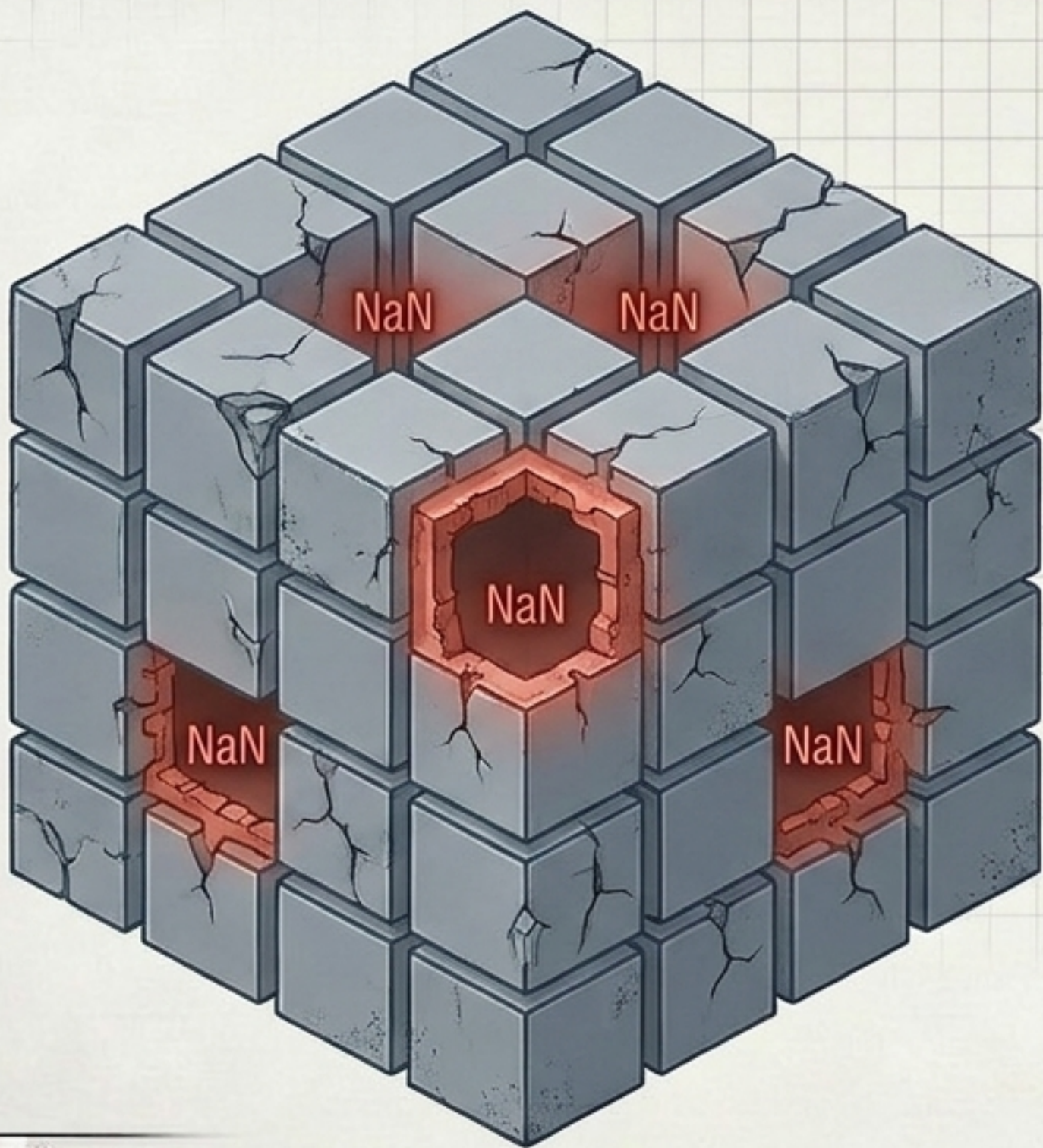
```
df.groupby('Department')['Salary'].mean()
```

Answers questions like:
“What is the average salary for each department?”

This pre-training analysis is vital for understanding feature distribution.



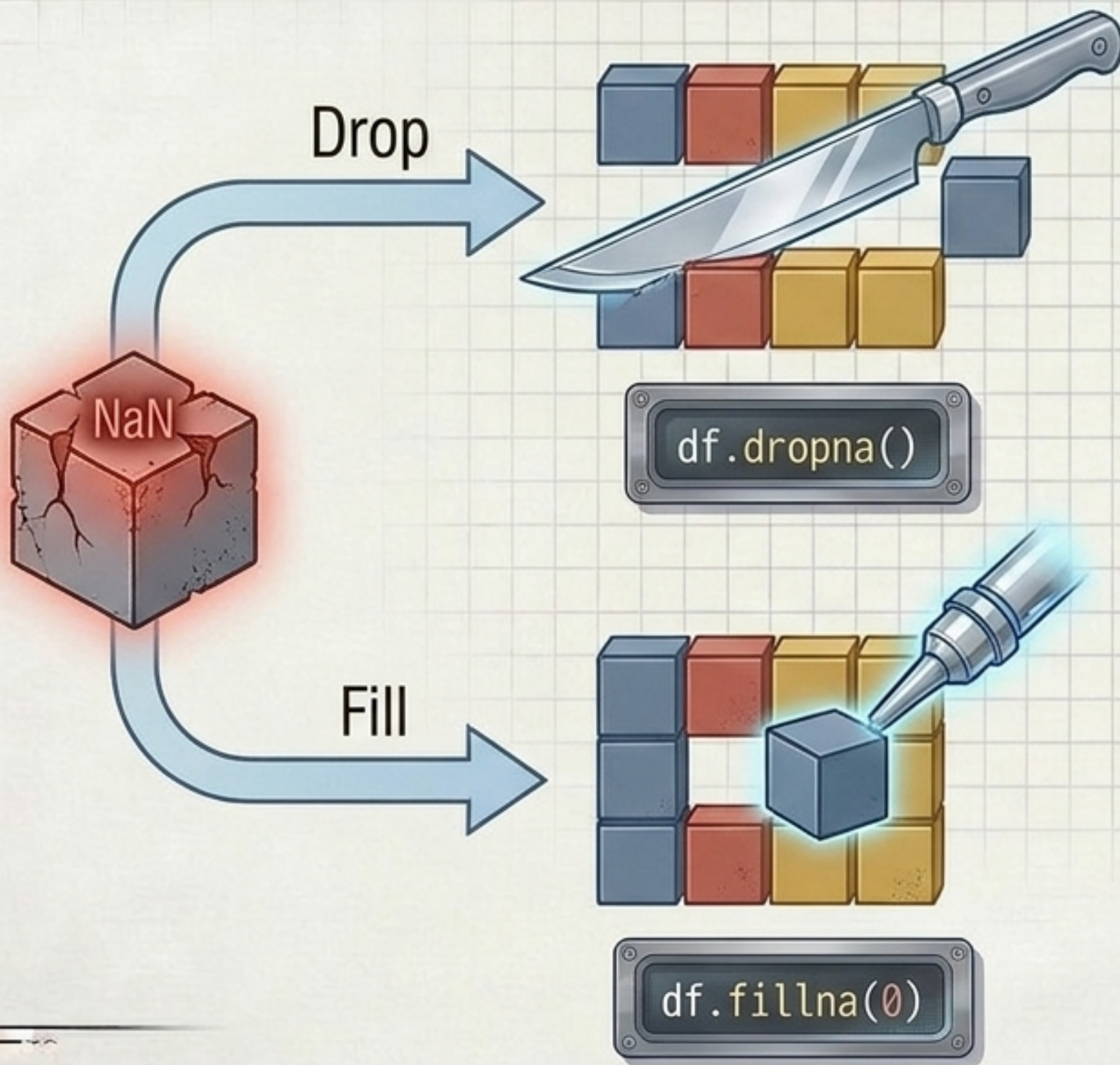
Missing Data: Real datasets are messy and riddled with missing values.



NaN (Not a Number) means the value is missing or unavailable.

```
# Locate missing values  
df.isnull()  
  
# Count total missing values per column  
df.isnull().sum()
```

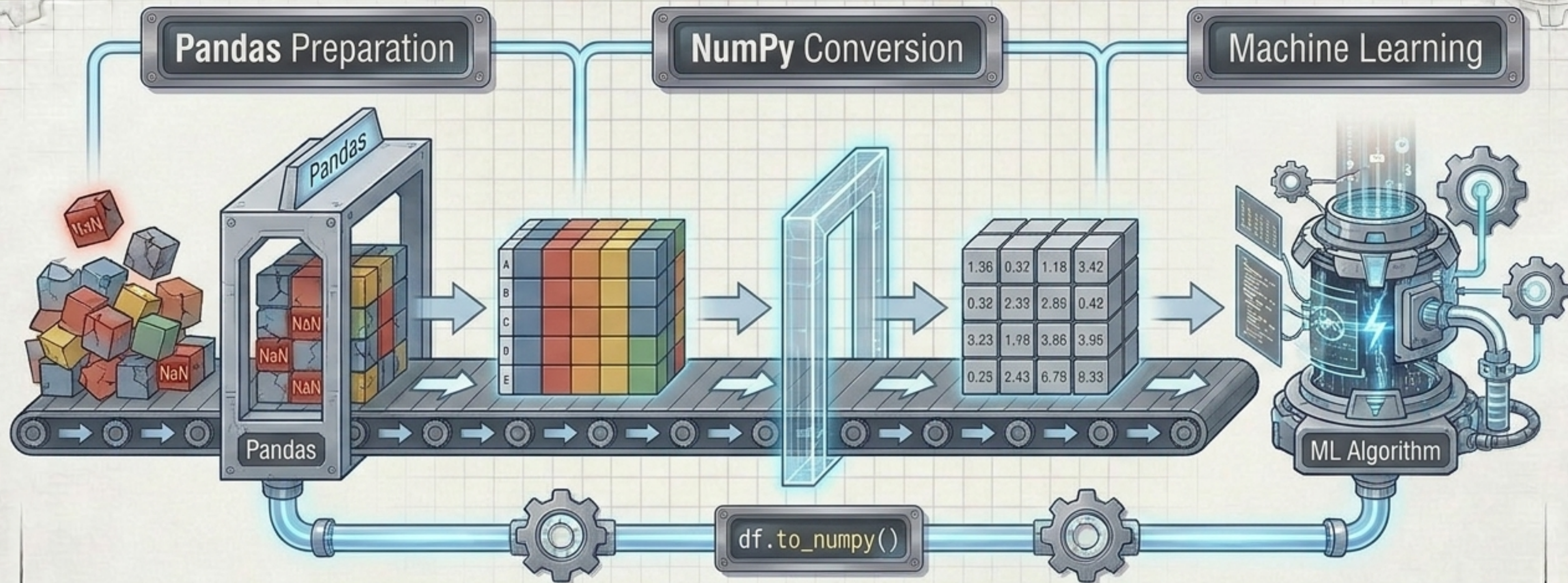

The Remediation Protocol: Never blindly delete or fill missing data.



Diagnostic Constraints (Drop vs. Fill depends on:)

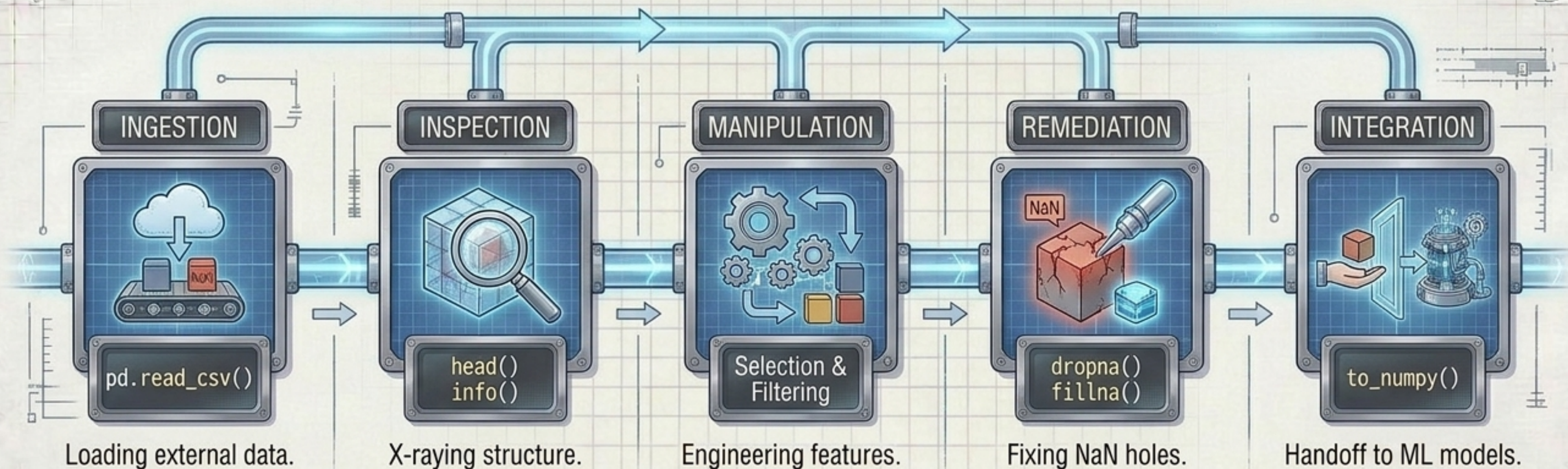
- ☒ 1. Why the value is missing.
- ☒ 2. The total size of the dataset.
- ☒ 3. The specific meaning of the feature.
- ☒ 4. The ultimate Machine Learning task.

The Ecosystem Handoff: From Pandas to Machine Learning.



Pandas handles the preparation. **NumPy** handles the numerical arrays. Once data is converted to raw NumPy arrays, it is officially a Machine Learning dataset.

The Complete Data Preparation Pipeline.



Pandas gives us the tools to turn messy, real-world data into something a Machine Learning algorithm can actually use.

Test your mental model.



The Refinery Test Data

Name	Score
Alice	90
Bob	75
Charlie	85

```
1 Create a dataframe `df`. Now mentally execute the pipeline:  
2 ☒ Display the first two rows.  
3   df.head(2)  
4 ☒ Find the average score.  
5   df['Score'].mean()  
6 ☒ Find students scoring above 80.  
7   df[df['Score'] > 80]  
8 ☒ Sort the data by score.  
9   df.sort_values('Score')  
5 ☒ Create a new 'Passed' column.  
10  df['Passed'] = True
```

Keep this deck on your digital desk. Master Pandas. Refine your data. Build better AI.